25 Runaway Reactions (content warning: #death, #eco-anxiety)

Why Are We Learning This?

Chemical reactors are designed to operate safely by balancing: Heat Generated and Heat Removed

However, under some conditions, heat generation can exceed heat removal. When this occurs, reactor temperature rises, reaction rates increase, and even more heat is produced.

Higher Temperature
↓
Faster Reaction
↓
More Heat Generated
↓
Higher Temperature

This positive feedback loop can lead to a runaway reaction.

Course Roadmap

Introduction to CRE
↓
General Balance Equation
↓
Common Reactor Models
(CSTR, Batch, PFR)
↓
Reaction Progress
(Conversion)
↓
Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)
↓
Rate Laws

(Kinetics)

↓

Equilibrium
(Reversible Reactions)

↓

Equilibrium Conversion
(Thermodynamic Limitations)

↓

One Reaction: Solving in terms of Reaction Progress

↓

One Reaction: Reactor Design
(CSTR, Batch, PFR, PBR)

↓

Catalytic Reactions

↓

Numerical Methods
(Python)

↓

Energy Balances

↓

One Reaction: Non-Isothermal Reactor Design

↓

Multiple Reactions

↓

►Reactor Safety◄

↓

Challenging the Well-Mixed Assumption
(Residence Time Distributions)

↓

Challenging the Elementary Rate Law Assumption
(Reaction Mechanisms)

Learning Objectives

By the end of this lesson, you should be able to:

- Define reaction runaway.

- Explain the temperature–rate feedback loop.
- Define the temperature of no return.
- Explain why latent reactions are dangerous.
- Identify factors that increase runaway risk.
- Explain why safety systems sometimes fail.
- Understand how calorimetry is used to identify hazardous conditions.
- Explain why scale-up requires detailed energy balances.

Big Idea

Heat Generated > Heat Removed
↓
Temperature Increases
↓
Reaction Accelerates
↓
More Heat Generated

If this cycle cannot be stopped, reaction runaway occurs.

- Reaction runaway is a risk whenever the heat generated > the heat removed in a chemical reactor.

- This is a risk, because chemical reactions give off *a lot* of heat, i.e., ΔH_{rxn}'s are large compared to other things (e.g., compare the heat of reaction for the formation of water versus the heat of vaporization of water from a prior lecture, where ΔH_{rxn} was 6x larger).

- The heat increases the temperature, which increases the rate constant, which increases the rate, which increases the heat generated $(-r_A)(-\Delta H_{rxn})$ term! So, once the heat generated exceeds the heat removed, it can be quite difficult -- if not impossible to control.

- Reaction runaway happens when it is no longer possible to control the heat generated term and hence temperature.

What Is Reaction Runaway?

Reaction runaway occurs when heat is generated faster than it can be removed from the reactor.

Physical Interpretation

Heat Generated > Heat Removed
↓
Temperature Rises
↓
Reaction Rate Increases
↓
Even More Heat Generated

The reactor becomes increasingly difficult to control because the reaction itself is accelerating the temperature increase.

Key Insight: Runaway Is a Feedback Problem

The reaction creates conditions that make itself proceed even faster.

- The temperature of no return is the temperature at which the rate constant is so large that reaction runaway cannot be prevented.

Am I Approaching Runaway?

Ask: As Temperature Increases, What Happens to the Reaction Rate?

If:

Higher Temperature
↓
Much Faster Reaction

then runaway risk increases.

Warning Signs: Temperature Increasing, Pressure Increasing, Heat Removal Becoming Less Effective

These are all indicators that the reactor may be approaching dangerous conditions.

What Is the Temperature of No Return?

The temperature of no return is the temperature at which the reaction rate becomes so large that runaway can no longer be prevented.

Physical Interpretation

Below Temperature of No Return
↓
Runaway May Still Be Stopped

Above Temperature of No Return
↓
Runaway Becomes Unavoidable

Key Insight

Temperature of No Return ≠ Boiling Point

Temperature of No Return ≠ Maximum Operating Temperature

It is a kinetic threshold where heat generation overwhelms all available heat-removal mechanisms.

- Some everyday examples of reaction runaway:

  - Cell phone battery fires (trigger warning: #death link) Li ion batteries involve an exothermic reaction … when a phone is poorly manufactured, overcharging or addition of heat (e.g., by leaving your phone in the sun) can result in reaction runaway.

Runaway Occurs Outside Chemical Plants Too

Runaway reactions are not limited to industrial reactors. Examples include: Lithium-Ion Battery Fires, Self-Heating Materials, Wildfire Feedback Loops

In each case:

Temperature Increase
↓
Process Accelerates
↓
Additional Heat Released

The same feedback mechanism appears in many systems.

- Global warming #eco-anxiety (link): Greenhouse emissions increase the Earth's temperature, which results in increased fires and releases of greenhouse gases that have been trapped in ice/permafrost, which contribute even more to temperature increases, which in turn cause more releases, and so on.

What Makes Runaway More Likely?

Runaway risk increases when one or more of the following are present:

High Heat Generation: Large Exothermic Heat of Reaction

Fast Reactions: Large Rate Constants

Strong Temperature Sensitivity: Large Activation Energy or rapidly increasing rates.

Poor Heat Removal: Weak Cooling System (Low Surface Area, Large Reactor Volume)

Limited Thermal Buffering: Few Inerts, Low Heat Capacity Materials

Unmodeled Reactions: may become important at elevated temperatures.

Engineering Concern

Model Predicts Recovery
↓

Temperature Gets Too High
↓
New Reactions Appear
↓
Actual Runaway Occurs

This is one reason why process safety analysis is so challenging.

- Examples from the textbook:

    - Example 12-7:
        - The combination of 1) a large processing rate (F_{A0} = 5 mol/min, which if you think about it is pretty large, especially for such a small (10 L) reactor), 2) coupled *exothermic* reactions, 3) high rate constants -- especially for reaction 1, which "turns over" at a rate of 40 per M per minute even at 300 K, 4) low activation barriers -- especially for reaction 1 of only 8 kJ/mol, and 5) the lack of any inerts with high heat capacities that could "absorb" some of the heat generated make it difficult to control the heat generated term.
        - 4 heat mitigation strategies are modeled: adiabatic, constant Ta, co-current heat exchange, counter-current heat exchange. If you examine the T versus V plots for each case, none of the strategies are able to prevent a large "spike" in the temperature: Constant Ta case, temperature spikes to 840 K; co-current heat exchange case, T spikes to 900 K; counter-current heat exchange, T spikes to 1100 K; adiabatic case, T spikes to 1600 K.
        - It appears that the temperature of no return in this example is ~ 390-400 K. This is where the temperature spike happens.
        - In each of these cases, the temperature eventually comes back down. This would suggest that, even though the temperature indeed gets really hot, the reaction doesn't actually run away. EXCEPT: We cannot know that from this example. This is because we have ONLY modeled the two reactions that we expect to occur under normal operation. If the temperature gets hot enough, other reactions could come into play.
        - These reactions aren't occurring under normal operation, likely because their rate constants are too small under normal operation,

likely due to large activation barriers. In other words, they are latent under normal operating conditions. However, as the temperature rises, these reactions could come into play. And if they are highly exothermic, they will lead to reaction runaway: i.e., as the new reactions proceed, the heat generated term gets larger, so large that the heat removal strategy can't compensate it, and the temperature will continue to increase rather than coming down.

    - Examples of latent reactions are: 1) new reactions between A, B, C, and D, 2) atomization or decomposition of A, B, C, and/or D (e.g., if one of these species is CH4 (g), you can imagine it decomposing to C (g) + 2H2 (g) or atomizing to C (g) + 4H (g) if the temperature gets super hot), 3) if this were a solvated reaction, A, B, C, and/or D could react with solvent, or the solvent could atomize or decompose!
- Problem 13-2_B:
    - Under normal operation, a molten ammonium nitrate/water solution was fed to the reactor. In the reactor, the solution decomposed to gaseous nitrous oxide and steam.
    - A "safety" feature was that the feed would be turned off.
    - However, the conversion of liquid reactants to gaseous products was a significant source of heat removal … without the feed, there was no heat removal!
- Example 13-2: Monsanto:
    - The heat exchanger failed for 10 minutes.
    - A 10 minute heat exchanger outage wouldn't normally be a problem. The heat exchanger had failed for ~ 10 minutes many times before, and it had been fine.
    - However, the reactor had recently been filled TOO MUCH. The normal design was to keep the reactor ~ 1/3 full of *liquid* (and in fact, a safety feature was that if the pressure got too high, a rupture disk would burst, which would decrease the pressure in the reactor, which would cause the water solvent to vaporize, which would serve as a heat removal strategy) but someone from the business side of the company had recently noticed this and ordered it be filled more completely, in order to increase the production rate (and hence the profit).

- Under this filling scenario, when the heat exchanger failed, even for 10 minutes, the $(-r_A)(-\Delta H_{rxn})$ term was too significant to be recoverable.
- AND THEN, the rupture disk also failed.
- So, the temperature got really hot, the pressure got really high, and the reactor exploded.

Why Safety Systems Must Match Actual Operation

Safety systems are designed for specific operating conditions. If the process changes:

Higher Production Rate, Larger Inventory, Different Filling Level

the original safety analysis may no longer be valid.

Key Lesson

Changing Throughput
↓
May Require New Safety Analysis

- Example 13-6: T2:
  - Heat exchanger failed, causing the "as designed" reactions to start to run out of control.
  - This resulted in an overproduction of H2 (g).
  - The production of H2 caused the pressure to increase.
  - As the pressure increased, so did the temperature.
  - As the temperature increased, the solvent started to atomize. This released even more H2 (g) and caused the pressure and temperature to increase further.
  - The reactor was equipped with a rupture disk, but it was not designed for the possibility that the solvent would atomize, so it was not helpful. Hence, the rate of H2 production far exceeded the venting rate, which ultimately lead to the reactor exploding.

Engineering Lesson: Accidents Rarely Have One Cause

Major industrial accidents often arise from multiple failures occurring simultaneously.

Typical Pattern

Small Problem
↓
Safety System Failure
↓
Unexpected Event
↓
Runaway

Key Insight

Multiple Independent Problems
↓
Combine Into Catastrophic Failure

- KEY POINT! Reaction runaway is often caused by reactions that are latent under normal operating conditions, but which come into play during a runaway scenario. These reactions have high activation barriers and hence negligibly small rates under normal operating conditions, but when the temperature approaches the temperature of no return, their rates are appreciable. These reactions are exothermic and hence contribute to the heat generated. Once they set in, they are nearly impossible to control. Hence, it is imperative to know 1) what reactions COULD occur given the chemicals involved in the reactor INCLUDING inerts, solvents, etc., (e.g., N_2 is often used as an inert, but it can atomize to $N_2 (g) \rightarrow 2N (g)$), 2) the temperature of no return.

What Are Latent Reactions?

Latent reactions are reactions that are essentially inactive during normal operation but become important at elevated temperatures.

Physical Interpretation

Normal Operating Temperature
↓
Rate ≈ 0

Elevated Temperature
↓
Rate Becomes Significant

These reactions often have large activation barriers and are therefore easy to overlook.

Why They Are Dangerous

Unexpected Reaction
↓
Additional Heat Generation
↓
Additional Temperature Rise
↓
More Unexpected Reactions

Once latent reactions begin, they can be extremely difficult to stop.

Safety Checklist: What Reactions Could Occur?

When evaluating reactor safety, ask:

Intended Reactions: What Reactions Are Designed to Occur?

Side Reactions: What Other Chemical Reactions Are Possible?

Decomposition Reactions: Could Any Species Decompose?

Solvent Reactions: Could Solvent Participate in Chemistry?

"Inert" Reactions: Could Any Inert Species React at High Temperature?

Every possible reaction pathway should be considered during safety analysis.

- Hence, even if the reaction you are carrying out is *endothermic*, reaction runaway is still a concern! Remember, it's not JUST the reactions that are part of the design equations, it's *every possible reaction given the chemicals involved*.

- The temperature of no return can be estimated in a calorimetry experiment. In this experiment, a very small amount of reaction mixture is heated in a adiabatic vessel that's closed to mass flow in or out until the temperature spikes. The reaction extinguishes quickly, since the amount of reaction mixture is very small.

How Do We Measure Runaway Risk?

One way to estimate the temperature of no return is through calorimetry experiments.

Procedure

Small Sample
↓
Adiabatic Environment
↓
Gradually Heat Sample
↓
Observe Temperature Response

Goal: Identify Temperature Where Self-Heating Accelerates Rapidly.

This temperature provides valuable information for reactor safety and process design.

- Scale up:
    - In senior design and maybe also in your career, you will estimate reactor design by looking into the literature and finding a residence time for a reaction carried out at "bench scale" (bench scale =, for example, a beaker) and then using that to estimate the reactor volume and heating/cooling requirements needed for an "industrial scale" reactor.

- Industrial scale reactors are often times A LOT LARGER than bench scale reactors, e.g., the size of a jacuzzi or even many, many times larger!
- The problem here is that the surface area to volume ratio is SMALLER for larger volume reactors.
- HENCE, the method of scale up using a residence time obtained from a bench scale experiment to determine the volume an heat exchanger requirements for an industrial scale reactor should only be used as AN ESTIMATE, which means you'll still need to solve the detailed material and energy balances to get an ACTUAL ANSWER.

Decision Guide: Why Is Scale-Up Difficult?

Students often assume: Large Reactor = Many Small Reactors

This is not true.

As Reactor Size Increases: Volume Increases Faster Than Surface Area

Consequence: Less Heat Removal Capacity Per Unit Volume

Engineering Impact

Bench-Scale Success
↓
Does Not Guarantee Industrial-Scale Success

A reactor that appears safe and well-controlled in a laboratory may behave very differently at industrial scale because heat removal becomes more difficult as reactor size increases.

Key Insight

Larger Reactor
↓
Lower Surface Area-to-Volume Ratio
↓
More Difficult Heat Removal
↓
Greater Runaway Risk

This is why bench-scale experiments should be viewed as a starting point for design rather than proof that a full-scale process will operate safely.

Key Insight

Residence Time from Bench Scale
↓
Useful First Estimate
↓
NOT a Substitute for Full Material and Energy Balances

Engineers must perform detailed material and energy balance calculations before finalizing an industrial reactor design.

- KEY POINT: using a residence time from a bench scale experiment to scale up a reactor IS ONLY AN ESTIMATE AND DOES NOT SUBSTITUTE FOR SOLVING THE FULL SET OF MATERIAL AND ENERGY BALANCE EQUATIONS.

Summary and Key Takeaways

Key Ideas

☑ Reaction runaway occurs when heat generation exceeds heat removal.

☑ Runaway is driven by positive feedback between temperature and reaction rate.

☑ The temperature of no return marks the threshold beyond which runaway becomes unavoidable.

☑ Latent reactions are often responsible for catastrophic runaway events.

☑ Safety analyses must consider all possible reactions, including decomposition reactions, solvent reactions, and reactions involving nominally inert species.

☑ Industrial accidents are often caused by multiple failures occurring simultaneously.

☑ Calorimetry can be used to estimate runaway hazards and identify temperatures of concern.

☑ Bench-scale performance does not guarantee industrial-scale safety.

Most Important Idea

Temperature Increases
↓
Reaction Accelerates
↓
More Heat Generated
↓
Temperature Increases

Reaction runaway is fundamentally a positive-feedback problem. The most effective way to prevent it is to understand all possible heat-generating processes before the reactor approaches the temperature of no return.